Riemann-Stieltjes Write Up

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March 2026

Definition 1. *Let $[a, b]$ be a closed interval. By a partition P of $[a, b]$ we mean a finite set of points $x_0, x_1, ..., x_n$, where $a = x_0 \leq x_1 \leq ... \leq x_n = b$.*
We write

$$\Delta x_i = x_i - x_{i-1} \ \ (i = 1, 2, ..., n) \tag{1}$$

Now suppose f is a bounded real function defined on $[a, b]$. Corresponding to each partition P of $[a, b]$ we put

$$M_i = \sup f(x) \ \ (x_{i-1} \leq x \leq x_i) \tag{2}$$

$$m_i = \inf f(x) \ \ (x_{i-1} \leq x \leq x_i) \tag{3}$$

$$U(P, f) = \sum_{i=1}^{n} M_i \Delta x_i \tag{4}$$

$$L(P, f) = \sum_{i=1}^{n} m_i \Delta x_i \tag{5}$$

If $\sup L(P, f) = \inf U(P, f)$, where $\sup$ and $\inf$ are taken over all partitions, then we write $f \in \mathscr{R}$ and we denote the common value as

$$\int_a^b f(x)dx. \tag{6}$$

Definition 2. *Let α be a monotonically increasing function on $[a, b]$. Correspondingly, we write*

$$\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1}) \ \ (i = 1, 2, ..., n). \tag{7}$$

Clearly $\Delta\alpha_i \geq 0$. For any real function f which is bounded on $[a, b]$ we put

$$U(P, f, \alpha) = \sum_{i=1}^{n} M_i \Delta\alpha_i \tag{8}$$

$$L(P, f, \alpha) = \sum_{i=1}^{n} m_i \Delta\alpha_i. \tag{9}$$

Where M_i and m_i are defined as in Definition 3. If $\sup L(P, f, \alpha) = \inf U(P, f, \alpha)$, where $\sup$ and $\inf$ are taken over all partitions, then we write $f \in \mathscr{R}(\alpha)$ and we denote the common value as

$$\int_a^b f d\alpha. \tag{10}$$

1 Theorems

Theorem 1. *Prove that if F and G are differentiable functions on $[a, b]$ and F' = $f \in \mathscr{R}$ and G' = $g \in \mathscr{R}$ then*

$$\int_a^b f(x)G(x)dx = F(b)G(b) - F(a)G(a) - \int_a^b F(x)g(x)dx. \tag{11}$$

Proof. Let H(x) = F(x)G(x). From the product rule it follows that $H'(x) = F'(x)G(x) + F(x)G'(x) = f(x)G(x) + F(x)g(x)$. We want to use the Fundamental Theorem of Calculus on H'(x) but before we can do that we must show that H'(x) is integrable. First we show that if f $\in \mathscr{R}$ and g $\in \mathscr{R}$, then fg $\in \mathscr{R}$.
Suppose that f $\in \mathscr{R}$ and g $\in \mathscr{R}$. Note that $\phi(t) = t^2$ is continuous everywhere. Thus we write $fg = \frac{(f+g)^2-(f-g)^2}{4}$. Since the sum and difference of two riemann integrable functions are riemann integrable, f+g and f-g are both riemann integrable. Since $\phi(f)$ is riemann integrable it follows that $(f+g)^2$ and $(f-g)^2$ are integrable. Finally, one more sum and a constant multiple yields that $fg = \frac{(f+g)^2-(f-g)^2}{4}$ is riemann integrable.
We now use the fact that since f and g are riemann integrable, it follows that F and G are continuous ans thus also riemann integrable. Therefore, $H'(x) = F'(x)G(x) + F(x)G'(x) = f(x)G(x) + F(x)g(x)$ is riemann integrable. Now that we have shown that $H'(x)$ is riemann integrable it follows from the Fundamental Theorem of Calculus that
$\int_a^b H'(x)dx = H(b) - H(a) = F(b)G(b) - F(a)G(a)$
$\implies \int_a^b f(x)G(x) + F(x)g(x)dx = F(b)G(b) - F(a)G(a)$
$\implies \int_a^b f(x)G(x)dx + \int_a^b F(x)g(x)dx = F(b)G(b) - F(a)G(a)$
$\implies \int_a^b f(x)G(x)dx = F(b)G(b) - F(a)G(a) - \int_a^b F(x)g(x)dx$ as desired. □

Theorem 2. *Evaluate the integral*

$$\int_a^b f d\alpha. \tag{12}$$

where $\alpha(x) = I(x)$ is the unit step function defined below

$$I(x) = \begin{cases} 0 & x < 0 \\ 1 & x \geq 0 \end{cases} \tag{13}$$

Proof. Consider a partition P on $[a, b]$ $a = x_0 < x_1 < ... < x_n = b$. If $0 \notin [x_i, x_{i+1}]$ then we have the following two cases: If $x_i > 0$ then $\Delta\alpha_i = \alpha(x_{i+1}) - \alpha(x_i) = 1 - 1 = 0$ and if $x_{i+1} < 0$ then $\Delta\alpha_i = \alpha(x_{i+1}) - \alpha(x_i) = 0 - 0 = 0$. In either case, any interval that does not contain 0 will add 0 to the upper sum and lower sum, so we only have to consider the interval $[x_k, x_{k+1}]$ such that $0 \in [x_k, x_{k+1}]$. For this interval, we note that $\Delta\alpha_k = \alpha(x_{k+1}) - \alpha(x_k) = 1 - 0 = 1$. Therefore on this interval, $U(P, f, \alpha) = L(P, f, \alpha) = f(0)(1)$ We choose 0 as the argument as we can make these partitions arbitrarily fine and thus for any number x that isn't 0 we can find a partition P such that x is not in the interval that also contains 0 since x $\neq 0 \iff \exists \epsilon > 0 s.t. |x - 0| \geq \epsilon$ and thus choose P such that for the interval $[x_i, x_{i+1}]$ s.t. $0 \in [x_i, x_{i+1}]$, $x_i > -\epsilon$ and $x_{i+1} < \epsilon$. Evaluating the integral in question we get that

$$\int_a^b f d\alpha = f(0) \tag{14}$$

if $a < 0$ and $b > 0$,

$$\int_a^b f d\alpha = -f(0) \tag{15}$$

if $a > 0$ and $b < 0$ (we take $\int_a^b f d\alpha = -\int_b^a f d\alpha$), and

$$\int_a^b f d\alpha = 0 \tag{16}$$

if $sgn(a)sgn(b) > 0$ (i.e. a and b have the same sign). Now we look at the case where 0 = a or 0 = b. In these cases, consider the interval $[0, x_1]$ and $[x_{n-1}, 0]$ for each of the cases respectively (all others go to 0 for the same reasons as above). Since the sum is forced to include the origin regardless of how small the intervals are, it follows that at 0, $\alpha(x) = 1$ and everywhere else $\alpha(x) = 0$. Thus $U(P, f, \alpha) \geq f(0)(1)$ while $L(P, f, \alpha) = f(0)(0)$. Thus $U(P, f, \alpha) - L(P, f, \alpha) \geq f(0)$ and therefore the integral in question does not exist if $f(0) \neq 0$. If $f(0) = 0$, the integral might exist. □

Theorem 3. *Suppose that $f(x) = 0$ for all irrational x, $f(x) = 1$ for all rational x. Then $f(x) \notin \mathscr{R}$ on $[a, b]$ for any $a < b$.*

Proof. Suppose that $f(x) = 0$ for all irrational x, $f(x) = 1$ for all rational x. Consider a partition P $\{x_0, x_1, ..., x_n\}$. From the density of $\mathbb{Q}$ in $\mathbb{R}$ and the irrationals being dense in $\mathbb{R}$(Proof: from density of $\mathbb{Q}$ in $\mathbb{R}$ it follows that for any two real numbers a and b there exists a rational number r such that $a - \sqrt{2} < r < b - \sqrt{2} \implies a < r + \sqrt{2} < b$. Since r is rational and $\sqrt{2}$ is irrational it follows that r + $\sqrt{2}$ is irrational), it follows that for any interval $[x_{i-1}, x_i]$ it follows that there exists an irrational number i $\in [x_{i-1}, x_i]$ and a rational number r $\in [x_{i-1}, x_i]$. Thus $M_n = 1$ and $m_n = 0$. Thus for any partition P, $U(P, f) > 0$ (as $M_n = 1$ and Δx is strictly positive) and $L(P, f) = 0$. Thus $\sup L(P, f) = 0$ and $\inf U(P, f) > 0$ and therefore $f(x) \notin \mathscr{R}$ on $[a, b]$ for any $a < b$. □

Theorem 4. *Consider the following two real-valued functions on* $[0, 1]$.

$$f(x) = \begin{cases} 1, x = \frac{1}{n} \text{ for some } n \in \mathbb{N} \\ 0, \text{otherwise} \end{cases} , g(x) = \begin{cases} n, x = \frac{1}{n} \text{ for some } n \in \mathbb{N} \\ 0, \text{otherwise} \end{cases} \tag{17}$$

Show (from the definition) that $f \in \mathscr{R}$ (i.e. f is Riemann-integrable), with $\int_0^1 f(x)dx = 0$, but $g \notin \mathscr{R}$.

Proof. Since between any two real numbers there exists an irrational number in between them it follows that $L(P, f) = 0$ for every partition P and thus $\sup L(P, f) = 0$. We want to show that $\inf U(P, f) = 0$. To do this we construct a sequence of partitions P_n in the following manner: We start with the interval $[0, \frac{\frac{1}{n}+\frac{1}{n+1}}{2}]$ which captures all of the numbers $\frac{1}{m}$ where $m > n$ and will have measure $(1)(\frac{\frac{1}{n}+\frac{1}{n+1}}{2}) = \frac{\frac{1}{n}+\frac{1}{n+1}}{2} = \frac{\frac{n+1}{n(n+1)}+\frac{n}{n(n+1)}}{2} = \frac{\frac{2n+1}{n(n+1)}}{2} = \frac{2n+1}{2n(n+1)} = \frac{2n+2-1}{2n(n+1)} = \frac{2(n+1)}{2n(n+1)} - \frac{1}{2n(n+1)} = \frac{1}{n} - \frac{1}{2n(n+1)}$. Now we consider the points $1/i, i = 1, ..., n$. For each one, we construct the interval $[\frac{1}{i} - \frac{1}{4n^2(n+1)}, \frac{1}{i} + \frac{1}{4n^2(n+1)}]$ unless i = 1 in which we construct $[1 - \frac{1}{2n^2(n+1)}, 1]$. Thus $U(P_n, f) = \frac{1}{n} - \frac{1}{2n(n+1)} + n(\frac{1}{2n^2(n+1)}) = \frac{1}{n} - \frac{1}{2n(n+1)} + \frac{1}{2n(n+1)} = \frac{1}{n}$. Since $L(P, f) \leq U(P, f)$, it follows that $0 \leq \inf U(P, f)$ and from our sequence of partitions it follows that $\inf U(P, f) \leq 0$ (as if it was the case that $\inf U(P, f) = \epsilon > 0$, we can always find a partition that had a smaller upper sum by choosing n such that $n > \frac{1}{\epsilon}$ and then choosing the partition P_n). Thus $\inf U(P, f) = 0$. Therefore, f $\in \mathscr{R}$ with $\int_0^1 f(x)dx = 0$. For g since g is not bounded (if $|g(x)| \leq M$ for all x and for some M choose a natural number $N > M$ and therefore $g(1/N) = N > M$), it immediately follows that g $\notin \mathscr{R}$. $\square$